Project Proposal

Project Proposal: Vehicle Financial Decision Engine

Group ID: 20

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Project Summary

One of the biggest purchases individuals make after buying a home is a car. This signals the importance of making the right decision and the consequences it may have down the line years after the purchase. One obvious metric the majority of the buyers focus on is the listed price of the vehicle they are interested in. However, while the listed price alone is a reasonable metric for buyers to consider, an often overlooked aspect of purchasing a vehicle is the long term financial ripple that may be hidden at the time of the purchase.

The financial ripple is a combination of financial aspects a vehicle purchase entails that are less apparent at the time of the purchase. One key financial component is the depreciation value of a vehicle in the region specific to the buyer. Vehicle depreciation is a reduction in the vehicle's market value over time, expressed as the difference between the original purchase price and the current resale value. A key feature of our platform is its geographic dimension. Vehicle depreciation does not behave uniformly across regions. A truck may hold its value exceptionally well in rural Northern California while depreciating faster in dense urban markets like San Francisco. Our platform allows users to visually compare depreciation curves across different regions, ensuring buyers understand how their specific location influences the long term value of their purchase. Along with that, interest rate and the insurance costs are crucial aspects of purchasing and owning a car that have lasting impact on the buyer. Essentially, our project aims to surface and highlight those pain points that buyers are often unaware of at the time of purchase and implement it into a system that buyers can use to make informed financial decisions.

We will be working with multiple datasets in order for this project to fulfill its promise. Datasets such as vehicle registration data, insurance data and vehicle sales data will be incorporated into our prediction models. Our project will use ML to assign each vehicle a depreciation rating out of 5 and the website will also let users visually see and compare the depreciation graphs among other competitors or choices. This rating is derived from a vehicle's historical value retention over a certain period relative to its original MSRP, weighted alongside reliability scores and regional market demand. A 5-star vehicle retains its value consistently over time and represents a financially sound purchase, while a 1-star vehicle signals rapid depreciation and higher long-term financial risk. As of now, CA will be our scope of initial market. However, we will design the project keeping scalability in mind to expand to other states.

Broader Impacts

This project will help a diverse group of buyers make financially informed decisions while purchasing a vehicle. First time buyers, student buyers with less financial capability, senior buyers who may be less informed on the current market can take advantage of our platform to make decisions that not only will be financially beneficial but also help them prepare for the painstaking negotiation and decision-making aspect of purchasing a car. On top of that, buyers will understand the conditions of the vehicle market, the resale trends and other miscellaneous nuances that can play a huge role in their finances. Through our project, buyers will be smarter and well equipped with the right strategy to save the most amount of money for them over time.

On the other hand, small dealerships can use our platform to adjust accordingly to market demand and depreciation value to compete with larger dealerships. Auto insurance premiums are partly tied to vehicle value. Regional depreciation data could inform more accurate premium pricing, particularly for smaller regional insurers who don't have large internal data science teams. Lastly, Uber, Lyft, and DoorDash drivers depend on their vehicles as income-generating assets. Understanding depreciation and total cost of ownership is directly tied to their profitability. Through our platform, they will be able to make financially smarter decisions that will result in a better economic impact.

Data Sources

National Association of Insurance Commissioners, *Auto Insurance Database Report*, Average Expenditure by State, 2019–2023

Reese, A. (2021). *Used cars dataset* [Data set]. Kaggle. https://www.kaggle.com/datasets/austinreese/craigslist-carstrucks-data

We will cite more datasets as we discover more appropriate datasets in the future

Expected Major Findings / Project Questions (List several questions/hypotheses/outcomes)

Our project will answer how geography within CA influences the depreciation of a vehicle and what factors of a vehicle have the most influence on its depreciation value. It will also address what combinations of factors(such as price, interest rate, location and demand) can output the best depreciation value. The project can also compare and contrast different cost of ownership over two or more vehicles with different depreciation trajectories. It can also demonstrate which vehicle segments have a more stable depreciation trajectory across different regions of CA. This project could debunk potential misconceptions, misassumptions of owning certain luxury vehicles that are financially less optimal.